

**EXACT STITCHED-SHEET, LOCAL-BOX, AND
COMPLEX-PHASE-TUBE POSITIVITY FOR A
HERMITIAN RANK-TWO SCALAR GATE**

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ABSTRACT. We study a scalar gate arising from a common-metric problem for sectorial matrix squares. After normalizing a balanced rank-two separator and a rank-one positive side in dimension three, the remaining complex Hermitian layer is a fully conjugated scalar gate involving Q and Q^2 . On one Schur-boundary branch we prove that the gate is a strictly convex cubic in a positive coupling scale and give a finite exact sign criterion. Rational Bernstein certificates then prove strict positivity on an explicit six-parameter shape box, including $3626/919 \leq t \leq 3726/919$, for every legal scale.

Independently, we certify six formula-distinct compact-ball sheets joined only at exact seams. A centered moving sheet is gate-positive through its closed breakpoint $X = 428905727/1858957100$. An affine- ω sheet reaches $X = 3/13$ with danger reserve $D \geq S/100$, and an inward- Z sheet continues to $X_0 = 83059/100000$. From that seam the recentered formula

$$Z_{\text{rec}} = Z_{\text{in}} + 2(X - X_0)$$

is strictly gate-positive on eight rational cells. The terminal value is $X_{\text{rec}}^* = 1019767/1040000$. Each recentered cell has 40/40 strict controls and complete rank-two legality for both lifts. At X_{rec}^* the $S = 0$ danger base vanishes, but an exact negative bound on the remaining affine term yields $D > 0$ for every physical $S > 0$. Immediately to the right, small positive S makes this particular chart illegal; this is not a negative raw gate or a maximality theorem.

A seam-preserving constant- Z lift then continues the strict gate certificate through $X = 497/500$ in two exact cells. A new cap shear

$$x = x_{\text{old}} - \frac{3}{2} \left(X - \frac{497}{500} \right), \quad y = y_{\text{old}}, \quad Z = Z_{\text{const}} + 2 \left(X - \frac{497}{500} \right)$$

splices at three algebraic levels and closes the charted interval through $X = 1$. Its 72/72 exact controls are strict. These stitched sheets remain local parameterized families, not a covering of the compact ball.

In a geometrically independent all-scale family we prove strict positivity on the stitched two-coordinate box

$$Z = \frac{1}{8}, \quad \frac{313}{500} \leq x \leq \frac{7}{10}, \quad |y| \leq \frac{1}{100},$$

for both signed lifts, every $0 < h \leq 1$, and every $\lambda > 0$. Four exact rational cells splice at the parameter, Hermitian, fully conjugated raw-gate, and complete boundary-control levels. Each cell has 531/531 strict exact three-variable Bernstein controls.

We also prove an all-scale high- Z complex-phase tube

$$0 < S \leq 1, \quad \lambda > 0, \quad 0 \leq x \leq \frac{1}{8}, \quad |y| \leq \frac{1}{64}, \quad z^2 = \frac{63}{64} - x^2,$$

for both signs of z , using 300 strictly positive exact controls. Independent no-import programs reconstruct every computational theorem from the original gate exactly; sampled nodes are only falsification checks. These local theorems do not prove the unrestricted complex gate, the common-metric assertion, the full compact ball, or an optimal fixed-lens spectral constant.

1. INTRODUCTION

Intersections of two disks and the associated sectorial numerical-range problems lead to angle-dependent spectral-set constants. The classical lens reduction and bounds are described by Beckermann and Crouzeix [2]; related sector estimates appear in [3], and uniform bounds for intersections of spherical disks appear in [1]. Further numerical-range estimates and perspectives can be found in [4, 5]. The present paper does not improve any of those constants. It isolates and certifies several local families in one matrix inequality that arises along a possible common-metric route.

Let $P, Q \succeq 0$ be finite-dimensional Hermitian matrices and put

$$(1) \quad A = \frac{(1+i)P + (1-i)Q}{2}, \quad B = A^2.$$

The common-metric problem asks whether there is a Hermitian matrix G_{met} satisfying

$$(2) \quad I \preceq G_{\text{met}} \preceq 2I, \quad B^* G_{\text{met}} + G_{\text{met}} B \succeq 0.$$

The unrestricted assertion in (2) is open in the setting relevant here. In dimension three, after fixing the balanced separator

$$(3) \quad W_0 = \frac{1}{2} \text{diag}(1, 1, 0)$$

and assuming that P has rank one, an exact arrowhead estimate leads to the scalar gate (12) below. Positivity of that scalar gate is a sufficient local step in the trace-dual problem. It is not equivalent to the existence of one metric for all separators.

The paper has ten mathematical results.

- (i) [Theorem 3.1](#) gives an exact scale-cubic reduction and a finite sign criterion on one Schur-boundary branch.
- (ii) [Theorem 4.1](#) proves the gate strictly positive on one explicit six-parameter shape box, with $3626/919 \leq t \leq 3726/919$, for the complete legal scale half-line. Counting that scale, this is a local seven-real-parameter chart.
- (iii) [Theorem 5.1](#) proves the same gate strictly positive on a different five-real-parameter moving-sheet family in a compact-ball chart on the closure $0 \leq X \leq X_*$, with a strict-danger theorem precisely for $0 \leq X < X_*$.
- (iv) [Theorem 6.1](#) proves a distinct affine- ω tilted chart on $1/5 \leq X \leq 3/13$ with $D \geq S/100 > 0$. It has an exact seam with the centered chart but does not enlarge that chart's parameter box.
- (v) [Theorem 7.1](#) proves a second, inward- Z tilted chart on the interval from $X = 3/13$ to $X = 83059/100000$. It has an exact parameter-by-parameter seam with the affine- ω chart and a rational bracket for the first loss of this chart's legality. It is a separate sheet, not a global continuation or maximality principle.

- (vi) [Theorem 8.1](#) proves a third formula-distinct compact-ball sheet from $X = 83059/100000$ to $X = 1019767/1040000$. It bypasses the preceding chart's legality root, includes its own exact danger-root endpoint, and stops only because that displayed recentering becomes illegal immediately to the right.
- (vii) [Theorem 9.1](#) proves a fourth continuation sheet from $X = X_{\text{rec}}^*$ through $X = 497/500$ by holding the recentering increment constant. Its two exact cells splice at $X = 99/100$.
- (viii) [Theorem 10.1](#) proves a fifth continuation sheet from $X = 497/500$ through $X = 1$ by a simultaneous rational shear of x and Z . It has an exact three-level seam with the constant- Z sheet.
- (ix) [Theorem 11.1](#) treats $Z = 1/8$. It proves an independent all-scale box in two shape coordinates on four exact cells from $x = 313/500$ through $x = 7/10$, including both signs of z .
- (x) [Theorem 12.1](#) proves an all-scale high- Z tube with two real shape parameters in the same compact-ball chart, including both signs of z .

The stitched compact-ball family consists of the centered, affine- ω , inward- Z , recentered- Z , constant- Z , and cap-shear sheets, joined only along their five exact seams. The all-scale Schur-boundary island, the stitched family, the $Z = 1/8$ local box, and the high- Z tube are separate local results. The terminology asserts neither a single global parameterization nor coverage of the regions between them. The all-scale island and the compact-ball theorems are parallel, not cumulative. They use different normalized rank-one sides,

$$(4) \quad p_A = (3/5, 0, 4/5)^T, \quad p_B = (1/\sqrt{6}, 0, \sqrt{5/6})^T,$$

and no equivalence, overlap, connectedness, or covering relation between their parameter domains is asserted. The affine- ω , inward- Z , recentered- Z , constant- Z , cap-shear, local-box, and high- Z theorems instead use the same p_B normalization. The affine chart meets the centered sheet only at its exact $X = 1/5$ seam, the inward chart meets the affine chart only at $X = 3/13$, the recentered chart meets the inward chart only at $X = 83059/100000$, the constant chart meets the recentered chart only at $X = 1019767/1040000$, and the cap chart meets the constant chart only at $X = 497/500$. These stitches establish connected unions, not one formula or a relation to Theorem A.

The dependency structure has a common first reduction and two separate final branches:

common-metric trace dual $\longrightarrow$ balanced rank-one-side scalar gate,
 scalar gate $\longrightarrow$ Schur boundary $\longrightarrow$ scale cubic $\longrightarrow$ Theorem A,
 scalar gate $\longrightarrow$ transverse compact ball,
 $\longrightarrow$ moving sheet $\longrightarrow$ Theorem B,
 $\longrightarrow$ affine- ω tilted sheet $\longrightarrow$ Theorem D,
 $\longrightarrow$ inward- Z tilted sheet $\longrightarrow$ Theorem E,
 $\longrightarrow$ recentered- Z sheet $\longrightarrow$ Theorem F,
 $\longrightarrow$ constant- Z sheet $\longrightarrow$ Theorem G,
 $\longrightarrow$ cap x -shear $\longrightarrow$ Theorem H,
 $\longrightarrow$ stitched $Z = 1/8$ local box $\longrightarrow$ Theorem I,
 $\longrightarrow$ high- Z phase tube $\longrightarrow$ Theorem C.

2. FROM THE COMMON METRIC TO THE SCALAR GATE

2.1. The trace-dual inequality. Write $\|\cdot\|_1$ for the trace norm. The following elementary separation identity fixes the coefficient in the dual inequality.

Proposition 2.1 (Finite-dimensional trace dual). *For fixed $B \in \mathbb{M}_n(\mathbb{C})$, (2) holds if and only if*

$$(5) \quad \|BW + WB^*\|_1 + 6 \operatorname{Re} \operatorname{tr}(BW) \geq 0 \quad \text{for every } W \succeq 0.$$

Proof. For $W \succeq 0$, set $H_W = BW + WB^*$. Cyclicity of the trace gives

$$\operatorname{tr}(W(B^*G_{\text{met}} + G_{\text{met}}B)) = \operatorname{tr}(G_{\text{met}}H_W).$$

Write $G_{\text{met}} = I + K$, where $0 \preceq K \preceq I$. For every Hermitian H , spectral decomposition gives

$$\max_{0 \preceq K \preceq I} \operatorname{tr}(KH) = \operatorname{tr}(H_+) = \frac{\|H\|_1 + \operatorname{tr} H}{2}.$$

Consequently

$$\max_{I \preceq G_{\text{met}} \preceq 2I} \operatorname{tr}(G_{\text{met}}H_W) = \frac{1}{2}\|H_W\|_1 + 3 \operatorname{Re} \operatorname{tr}(BW).$$

The compact convex set $\{B^*G_{\text{met}} + G_{\text{met}}B : I \preceq G_{\text{met}} \preceq 2I\}$ meets the positive semidefinite cone exactly when no positive semidefinite separator makes this maximum negative. Multiplication by two proves the assertion. $\square$

2.2. The first mixed-state compression. Let $W \succeq 0$ have rank two in dimension three, let $\mathcal{S} = \text{ran } W$, and decompose relative to $\mathcal{S} \oplus \mathcal{S}^\perp$. If

$$C = P_{\mathcal{S}}B|_{\mathcal{S}}, \quad r = P_{\mathcal{S}^\perp}B|_{\mathcal{S}},$$

then

$$(6) \quad BW + WB^* = \begin{pmatrix} CW + WC^* & Wr^* \\ rW & 0 \end{pmatrix}.$$

We shall use the following exact arrowhead estimate. A proof is included in [Section A](#).

Lemma 2.2 (Arrowhead trace norm). *For $H_0 = H_0^* \in \mathbb{M}_2(\mathbb{C})$ and $y \in \mathbb{C}^2$,*

$$(7) \quad \left\| \begin{pmatrix} H_0 & y \\ y^* & 0 \end{pmatrix} \right\|_1^2 \geq \|H_0\|_1^2 + 4\|y\|^2.$$

Put $H_0 = CW + WC^*$, $y = Wr^*$, and $\alpha = \text{Re tr}(CW) = \text{tr}(H_0)/2$. If $\alpha \geq 0$, the trace-dual inequality is automatic. If $\alpha < 0$, Lemma 2.2 shows that

$$(8) \quad \|H_0\|_1^2 + 4\|y\|^2 \geq 36\alpha^2$$

is sufficient for (5); moreover, every strict failure of (5) forces the strict reverse of (8).

Now impose (3), let $P = pp^*$ with

$$(9) \quad p = (a, 0, c)^T, \quad a, c > 0, \quad a^2 + c^2 = 1,$$

and let the other positive side Q be arbitrary positive semidefinite. Put $\xi = Qp$. In the dangerous chart $\text{Re } \xi_1 < 0$, direct multiplication of $B = A^2$ gives

$$(10) \quad H_0 = \begin{pmatrix} a \text{Re } \xi_1 & a \overline{\xi_2}/2 \\ a \xi_2/2 & 0 \end{pmatrix},$$

and the leakage row has entries

$$(11) \quad \begin{aligned} 2r_1 &= c \overline{\xi_1} + a \xi_3 + i\{ac - (Q^2)_{31}\}, \\ 2r_2 &= c \overline{\xi_2} - i(Q^2)_{32}. \end{aligned}$$

Since the determinant of H_0 is nonpositive,

$$\|H_0\|_1^2 = a^2(\text{Re } \xi_1)^2 + a^2|\xi_2|^2.$$

Thus (8) is precisely $\mathcal{G}(Q) \geq 0$, where

$$(12) \quad \boxed{\begin{aligned} \mathcal{G}(Q) &= a^2|\xi_2|^2 + \frac{1}{4} \left| c \overline{\xi_1} + a \xi_3 + i\{ac - (Q^2)_{31}\} \right|^2 \\ &\quad + \frac{1}{4} \left| c \overline{\xi_2} - i(Q^2)_{32} \right|^2 - 8a^2(\text{Re } \xi_1)^2. \end{aligned}}$$

Both entries of Q^2 and all conjugations in (12) are part of the definition.

For the rest of the paper we use the single normalization

$$(13) \quad \Gamma(Q) := 4\mathcal{G}(Q).$$

Accordingly, the scale-cubic polynomial below is $\Delta^2\Gamma = 4\Delta^2\mathcal{G}$, while the compact-ball normalization is $36\Gamma = 144\mathcal{G}$. The symbol G_{met} is reserved for the metric in (2); it is never used for the scalar gate.

Remark 2.3 (Logical scope). Theorems proving $\mathcal{G}(Q) \geq 0$ verify the sufficient scalar condition (8) for the balanced separator and the stated rank-one side. They do not select one G_{met} valid for all separators and therefore do not prove (2).

3. THE EXACT SCHUR-BOUNDARY SCALE CUBIC

After using the middle-coordinate phase to make Q_{32} real and nonnegative, write

$$(14) \quad Q = \begin{pmatrix} r & u+iv & z+iw \\ u-iv & q & s \\ z-iw & s & t \end{pmatrix} \succeq 0, \quad s > 0,$$

with all displayed scalar parameters real. In the active order $\{1, 3\} \mid \{2\}$, set

$$(15) \quad R = \begin{pmatrix} r & z+iw \\ z-iw & t \end{pmatrix}, \quad h = \begin{pmatrix} u+iv \\ s \end{pmatrix}, \quad \Delta = rt - z^2 - w^2.$$

On the strict active chart $R \succ 0$, positivity of Q is equivalent to

$$(16) \quad q \geq q_{\min} := h^* R^{-1} h = \frac{t(u^2 + v^2) + rs^2 - 2s(uz + vw)}{\Delta}.$$

The only q -dependent square in (12) is

$$(17) \quad \left| c\bar{\xi}_2 - i(Q^2)_{32} \right|^2 = \{c(au + cs) + zv - wu\}^2 + s^2(q - q_g)^2,$$

where

$$(18) \quad q_g = -t + \frac{acv - zu - wv}{s}.$$

Hence the constrained scalar minimizer is $\max\{q_{\min}, q_g\}$. We study the Schur-boundary branch $q = q_{\min} > q_g$.

Separate the coupling direction from its scale by writing

$$(19) \quad u = sk, \quad v = s\ell, \quad T = s^2 > 0.$$

Define

$$(20) \quad \begin{aligned} n &= t(k^2 + \ell^2) + r - 2(kz + \ell w), \\ g &= ac\ell - zk - w\ell - t, \\ x_0 &= ar + cz, \quad L = ak + c, \\ A_0 &= ac(r + t) + z - w(r + t), \\ B_0 &= ac - w - z(r + t), \quad C_0^\flat = cL + z\ell - wk. \end{aligned}$$

Then $n > 0$, $q_{\min} = Tn/\Delta$, and the strict branch is the half-line

$$(21) \quad T > T_L, \quad T_L = \begin{cases} 0, & g \leq 0, \\ \Delta g/n, & g > 0. \end{cases}$$

Theorem 3.1 (Scale-cubic criterion). *Assume (9), (14)–(19), $R \succ 0$, and $x_0 < 0$. On the closure $T \geq T_L$ of the branch (21),*

$$(22) \quad \Delta^2 \Gamma(Q) = P(T) = C_0 + C_1 T + C_2 T^2 + n^2 T^3,$$

where

$$(23) \quad C_0 = \Delta^2(A_0^2 + B_0^2 - 32a^2x_0^2),$$

$$(24) \quad C_1 = \Delta^2[-2(A_0\ell + B_0k) + 4a^2(L^2 + a^2\ell^2) + (C_0^\flat)^2 + g^2],$$

$$(25) \quad C_2 = \Delta^2(k^2 + \ell^2) - 2\Delta gn.$$

The polynomial P is strictly convex on $[T_L, \infty)$. If

$$P(T_L + S) = g_0 + g_1 S + g_2 S^2 + n^2 S^3,$$

then $g_2 > 0$ and

$$(26) \quad P \geq 0 \text{ on } [T_L, \infty) \iff g_0 \geq 0 \text{ and } [g_1 \geq 0 \text{ or } (g_1 < 0 \text{ and } \mathfrak{D} \leq 0)],$$

where

$$(27) \quad \mathfrak{D} = 18n^2 g_2 g_1 g_0 - 4g_2^3 g_0 + g_2^2 g_1^2 - 4n^2 g_1^3 - 27n^4 g_0^2$$

is the cubic discriminant.

Proof. At $q = q_{\min}$, direct expansion of (12) gives

$$\begin{aligned} P(T) = \Delta^2 & \left[(A_0 - T\ell)^2 + (B_0 - Tk)^2 - 32a^2x_0^2 \right. \\ & \left. + T\{4a^2(L^2 + a^2\ell^2) + (C_0^\flat)^2\} \right] + T(Tn - \Delta g)^2, \end{aligned}$$

which is (22)–(25). Moreover

$$P''(T) = 2(C_2 + 3n^2 T).$$

If $g \leq 0$, then $C_2 > 0$; if $g > 0$, then

$$C_2 + 3n^2 T_L = \Delta^2(k^2 + \ell^2) + \Delta gn > 0.$$

Thus P is strictly convex on the legal half-line. After shifting by T_L , if $g_1 \geq 0$ the cubic is increasing. If $g_1 < 0$, its derivative has exactly one positive zero, the unique minimum. With $g_0 \geq 0$, that minimum is negative, zero, or positive according as the cubic has two positive simple roots, a positive double root, or no positive root. These three alternatives are exactly $\mathfrak{D} > 0$, $\mathfrak{D} = 0$, and $\mathfrak{D} < 0$. In the endpoint case $g_0 = 0$ and $g_1 < 0$, one has $\mathfrak{D} = g_1^2(g_2^2 - 4n^2 g_1) > 0$, so the criterion correctly rejects the immediate descent from zero. This proves (26). $\square$

4. THE ALL-SCALE POSITIVITY ISLAND

We now fix $p = p_A$ from (4). For

$$(28) \quad \begin{aligned} 8 \leq \ell \leq 10, \quad & -\frac{31}{100} \leq w \leq -\frac{29}{100}, \\ -\frac{301}{100} \leq k \leq -\frac{299}{100}, \quad & -\frac{101}{100} \leq z \leq -\frac{99}{100}, \\ \frac{99}{100} \leq r \leq \frac{103}{100}, \quad & \frac{3626}{919} \leq t \leq \frac{3726}{919}, \end{aligned}$$

use (20) with $a = 3/5, c = 4/5$ and define, for $T > 0$,

$$(29) \quad Q_A(T) = \begin{pmatrix} r & \sqrt{T}(k + i\ell) & z + iw \\ \sqrt{T}(k - i\ell) & Tn/\Delta & \sqrt{T} \\ z - iw & \sqrt{T} & t \end{pmatrix}.$$

Theorem 4.1 (All-scale island). *For every parameter choice in (28), $\Delta, n > 0$, $Q_A(T)$ is positive semidefinite of rank two, and $\text{Re}(Q_A(T)p_A)_1 < 0$. On the complete strict branch*

$$(30) \quad T > 0 \quad (g \leq 0), \quad T > \Delta g/n \quad (g > 0),$$

one has

$$\mathcal{G}(Q_A(T)) > 0.$$

The cleared polynomial remains strictly positive at the closure endpoint $T = T_L$. In particular, equality does not occur in the stated legal family.

4.1. Exact sign certificate. For $g > 0$, define the two essential polynomial numerators

$$(31) \quad N_0 = 15625n^2P(T_L), \quad N_1 = 625nP'(T_L).$$

After denominators are cleared, both are polynomials in the six box variables. At $t = 4$, split the r interval into

$$(32) \quad [99/100, 101/100] \cup [101/100, 103/100].$$

Exact tensor Bernstein conversion proves that each of

$$(33) \quad -\text{Re}(Q_A p_A)_1, \quad \Delta, \quad n, \quad C_0, \quad C_1, \quad N_0, \quad N_1$$

has strictly positive controls on both five-dimensional cells. Each cell contains 40,595 theorem-relevant controls, with the degree and count distribution shown in Table 1.

To open the t coordinate without constructing a six-dimensional tensor, let f be any polynomial in (33). If m_f is its least center control and M_f is the freshly recomputed exact monomial majorant for $|\partial_t f|$ on the full box $3626/919 \leq t \leq 3726/919$, then

$$(34) \quad f(t) \geq m_f - |t - 4|M_f \geq m_f - 50M_f/919.$$

TABLE 1. Exact center-cell control counts. The same degree pattern is reconstructed independently on both r cells.

quantity	tensor degree	controls per cell
danger margin	$(0, 0, 0, 1, 1)$	4
Δ	$(0, 2, 0, 2, 1)$	18
n	$(2, 1, 2, 1, 1)$	72
C_0	$(0, 6, 0, 6, 4)$	245
C_1	$(2, 6, 2, 6, 3)$	1,764
N_0	$(5, 10, 5, 10, 6)$	30,492
N_1	$(4, 7, 4, 7, 4)$	8,000

All seven final reserves are positive. The smaller reserve across the two r cells is recorded exactly in [Section B](#). This bridge avoids the direct 9,975,375-control six-dimensional tensor. Among the fourteen cell/sign pairs, the unique finite ratio bottleneck is the lower- r value for N_0 :

$$\rho_{\text{wide}} = \frac{22832956414652720419288572875925058067205119308096}{419658897924777012091564281675537127592865447699025},$$

$$\rho_{\text{wide}} > \frac{50}{919}.$$

Thus the current half-width 50/919 is below the exact reserve ceiling on the box where the derivative majorants were certified. No larger half-width is claimed. The older eight-sign audit also tracked a discriminant numerator N_D , but N_D is absent from both terminal branches below.

Proof of [Theorem 4.1](#). The positive Δ, n controls imply that the active block is positive definite. The middle diagonal entry in (29) equals the exact Schur value Tn/Δ , so $Q_A(T) \succeq 0$ has rank two. The danger control is strictly positive after sign reversal. Thus (30) is precisely the branch in [Theorem 3.1](#).

If $g \leq 0$, positivity of C_0, C_1 and the identity

$$C_2 = \Delta^2(k^2 + \ell^2) - 2\Delta gn > 0$$

make every coefficient of $\mathbf{P}$ positive. If $g > 0$, positivity of N_0, N_1 gives $\mathbf{P}(T_L) > 0$ and $\mathbf{P}'(T_L) > 0$, while

$$C_2 + 3n^2T_L = \Delta^2(k^2 + \ell^2) + \Delta gn > 0.$$

Hence $\mathbf{P}$ increases strictly from a positive endpoint. Since $\Delta > 0$ and $\mathbf{P} = 4\Delta^2\mathcal{G}$, the original gate is strictly positive. $\square$

The family is genuinely complex. For example, the audited endpoint choice

$$(\ell, w, k, z, r, t, T) = (9, -31/100, -301/100, -101/100, 103/100, 3726/919, 1)$$

has opposite-sign imaginary coordinates and

$$\begin{aligned}\mathcal{G}(Q) &= \frac{60746927109990602200372569492531}{16695493268772709728100000000} > 0, \\ 4\{\mathcal{G}(Q) - \mathcal{G}(\operatorname{Re} Q)\} &= -\frac{184269381654495888525183}{118744947221210000000} < 0.\end{aligned}$$

Thus the proof is not a real-part monotonicity argument.

5. THE POSITIVE- Z MOVING-SHEET ISLAND

We now switch to the different normalized vector $p = p_B$ in (4). Put

$$(35) \quad \zeta = \frac{4+3i}{5}, \quad n_h = (c\sqrt{1-h^2}, h, a\sqrt{1-h^2}\zeta)^T,$$

where $a = 1/\sqrt{6}$, $c = \sqrt{5/6}$, and define

$$(36) \quad r_0 = (-a, 0, c\zeta)^T, \quad f_h = (-ch, \sqrt{1-h^2}, -ah\zeta)^T, \quad U_h = (r_0 \ f_h).$$

The columns of U_h , together with n_h , are orthonormal. For $h > 0$, $\lambda > 0$ and $j, \kappa, \ell \in \mathbb{R}$, put

$$(37) \quad H = U_h \begin{pmatrix} h^2 & h(j+i\kappa) \\ h(j-i\kappa) & j^2 + \kappa^2 + \ell^2 \end{pmatrix} U_h^*, \quad Q = \lambda H.$$

The compression determinant is $h^2\ell^2$, so Q is positive semidefinite and has rank two when $\ell \neq 0$.

Introduce

$$(38) \quad j = \frac{1+5x}{3\sqrt{5}}, \quad \kappa = \frac{-3+5y}{3\sqrt{5}}, \quad \ell = \frac{\sqrt{5}}{3}z, \quad S = h^2, \quad Z = z^2.$$

Direct multiplication gives

$$(39) \quad \operatorname{Re}(Qp_B)_1 = \frac{5\lambda S\sqrt{6}}{36}(x^2 + y^2 + Z - 1).$$

Thus strict danger is exactly the open unit ball.

The compact normalization is transparent from (12). With $\xi = Hp_B$, define

$$(40) \quad \begin{aligned}\mathcal{B}_0 &= (0, iac, 0)^T, \\ \mathcal{L} &= (2a\xi_2, c\bar{\xi}_1 + a\xi_3, c\bar{\xi}_2)^T, \\ \mathcal{M} &= (0, -i(H^2)_{31}, -i(H^2)_{32})^T, \quad d = \operatorname{Re} \xi_1.\end{aligned}$$

Then, with the global normalization (13),

$$(41) \quad \Gamma(\lambda H) = \|\mathcal{B}_0 + \lambda\mathcal{L} + \lambda^2\mathcal{M}\|^2 - 32a^2\lambda^2d^2.$$

This is a real quartic in λ , and exact cancellation of square roots gives

$$(42) \quad 36\Gamma(\lambda H) = 5 + S \sum_{m=1}^4 \lambda^m F_m(S, x, y, z),$$

where $F_m \in \mathbb{Q}[S, x, y, z]$, $\deg_S F_m \leq m - 1$, and $\deg_{x,y,z} F_m \leq 2m$. The sign of the full compact-ball quartic remains open.

5.1. The moving-sheet map. Let

$$(43) \quad 0 < S \leq \frac{1}{10000}, \quad 0 \leq X \leq X_*, \quad |M| \leq \frac{1}{1000}, \quad |\omega|, |\nu| \leq \frac{1}{100}.$$

Here and below

$$(44) \quad X_* := \frac{428905727}{1858957100}.$$

Set $\mu = 1 + M$ and define

$$(45) \quad \begin{aligned} \lambda &= \mu/S, \\ y_0 &= \frac{12}{25\mu} + \nu, \\ w_c &= -\frac{3(5MX - 15M + 5X - 6)}{25\mu}, \\ Y &= Sy_0, \quad W_{\text{sh}} = S(w_c + \omega), \\ Z &= 3X - X^2 - Y^2 + W_{\text{sh}}, \\ x &= -\frac{1}{5} + X, \quad y = \frac{3}{5} + Y. \end{aligned}$$

Theorem 5.1 (Positive- Z moving-sheet island). *For every parameter choice in (43), the map (45) satisfies*

$$(46) \quad 0 < Z < 1, \quad x^2 + y^2 + Z \leq 1, \quad \lambda > 0.$$

Choose either sign of z with $z^2 = Z$, and construct $Q_B = \lambda H$ from (37)–(38), explicitly using $h = \sqrt{S}$ and $\ell = \sqrt{5}z/3$. Then Q_B is positive semidefinite of rank two and

$$\mathcal{G}(Q_B) > 0.$$

For $0 \leq X < X_$ it lies in the strict dangerous chart. At $X = X_*$, danger is nonnegative and vanishes at exactly*

$$(47) \quad S = \frac{1}{10000}, \quad M = \frac{1}{1000}, \quad \omega = \nu = \frac{1}{100}.$$

Thus the uniform strict-danger domain is half-open even though the gate certificate is strict on its closure. The theorem does not allow arbitrary normalized scale: it imposes $\lambda S = 1 + M \in [999/1000, 1001/1000]$.

Legality. Exact rational interval bounds give

$$(48) \quad \frac{46999}{100100} \leq y_0 \leq \frac{16333}{33300}, \quad 0 < Y \leq \frac{16333}{333000000}.$$

The frozen predecessor cells through $X = 1/5$ prove $0 < Z < 1$ directly. On the terminal cell $1/5 \leq X \leq X_*$, independent interval envelopes give

$$(49) \quad \frac{14}{25} < Z \leq \frac{220824455717308720521}{345572149964041000000} < 1.$$

The identity

$$x^2 + y^2 + Z = \frac{2}{5} + \frac{13}{5}X + \frac{6}{5}Y + W_{\text{sh}}$$

gives, on the terminal cell,

$$(50) \quad 1 - x^2 - y^2 - Z = \frac{3}{5} - \frac{13}{5}X - S\mathcal{B}, \quad \mathcal{B} = \frac{9}{5} + \omega + \frac{6}{5}\nu - \frac{63}{125(1+M)} - \frac{3}{5}X.$$

There $\mathcal{B} \geq 585533340809/515860595250 > 0$. Its exact partial derivatives show that the minimum of (50) is at the corner (47); at that corner it is

$$(51) \quad -\frac{1858957100X - 428905727}{715000000}.$$

All monotonicities are strict. Hence danger is positive uniformly for $X < X_*$ and is zero at exactly the stated endpoint corner. The predecessor cells are strictly dangerous by their independently reconstructed reserves. Finally, the support compression in (37) has determinant

$$\frac{5}{9}SZ > 0.$$

Its leading entry is $S > 0$, so it is positive definite and Q_B has rank two. This proves the legality and sharp uniform strict-danger assertions. $\square$

5.2. The exact moving-sheet certificate. Let

$$(52) \quad \mathcal{N} = S^3(36\Gamma(Q_B)), \quad D = 25^8(1+M)^8, \quad \hat{Q} = D\mathcal{N}/S^2.$$

The positive denominator and the factor S^2 are exact: after the map (45), $D\mathcal{N}$ is divisible by S^2 . The quotient has exactly twenty (S, X) monomials and bidegree $(5, 4)$. Its constant layer is zero, and its first layer is

$$(53) \quad H_1 = 25^8 M^2 S(1+M)^8 (5M^2 + 14M + 14) \geq 0.$$

After removing H_1 , the homogeneous quadratic layer is

$$(54) \quad H_2 = a_2 S^2 + b_2 SX + c_2 X^2.$$

Exact centered-monomial bounds on the (M, ω, ν) box prove

$$(55) \quad a_2, b_2, c_2 \geq c_*, \quad c_* = \frac{2564950982194530478444050838857341987999}{127401984000000000000000000000} > 0.$$

Hence

$$(56) \quad H_2 \geq c_*(S^2 + SX + X^2).$$

Group every remaining monomial according to its excess total (S, X) degree $d = 1, \dots, 5$. Exact absolute centered-monomial bounds give constants $B_d > 0$, reconstructed from 947 parameter monomials, such that the absolute remainder is bounded by the corresponding sum. The five exact B_d are recorded in [Section B](#).

On the predecessor cell $0 \leq X \leq 3/10000$ there are two projective charts. If $X \leq S$, set $X = S\rho$ with $0 \leq \rho \leq 1$. After division by S^2 , (56) supplies the reserve c_* , while the higher remainder is at most $\sum_{d=1}^5 B_d S^d$. At $S \leq 10^{-4}$

the exact remaining margin is the positive rational m_{C0} in [Section B](#). This chart includes $X = 0$ without division by X . If $S \leq X$, set $S = X\sigma$ with $0 \leq \sigma \leq 1$; after division by X^2 , the predecessor remainder has exact margin $m_{C1} > 0$. The charts overlap at $X = S$.

On the adjacent cell $3/10000 \leq X \leq 1/31$, one has $0 < S \leq 1/10000 < 3/10000 \leq X$, so the same substitution $S = X\sigma$, now with $0 < \sigma < 1$, is lossless. Regenerating all 947 centered parameter monomials gives the strict margin $m_{\text{adj}} > 0$ in [Section B](#). On the next cell $1/31 \leq X \leq 1/30$, retain the sharper bound $\sigma \leq 31/10000$ term by term; the resulting strict margin is $m_{\text{next}} > 0$. Finally, on $1/30 \leq X \leq 1/24$, the lossless bound $\sigma \leq 3/1000$ gives $m_{\text{rec}} > 0$. Each cell regenerates the twenty (S, X) coefficients and all 947 centered parameter monomials. On the cross-cap cell $1/24 \leq X \leq 1/23$, use $\sigma \leq 3/1250$ and retain $T = Z - 1/8$ only as a recentered coordinate: the proof never divides by T . The resulting strict margin is $m_{\text{cross}} > 0$. On the new cell $1/23 \leq X \leq 1/22$, the same lossless order gives $\sigma \leq 23/10000$. Retaining every exact power of σ in the sixteen higher terms and regenerating all 947 centered monomials gives the strict margin $m_{22} > 0$ in [Section B](#). The six closed cells through $X = 1/22$ meet at $X = 3/10000$, $1/31$, $1/30$, $1/24$, and $1/23$. On $1/22 \leq X \leq 1/21$, the lossless order $\sigma \leq 11/5000$ gives the independently reconstructed strict margin $m_{21} > 0$. On $1/21 \leq X \leq 1/20$, the lossless order $\sigma \leq 21/10000$ gives the independently reconstructed strict margin $m_{20} > 0$. These last two closed cells meet the predecessor exactly at $X = 1/22$ and each other exactly at $X = 1/21$; no limit argument is used. On the new cell $1/20 \leq X \leq 1/19$, put

$$u = 380(X - 1/20), \quad 0 \leq u \leq 1,$$

so that $u = 0$ is exactly the preceding seam and $u = 1$ is the new endpoint. Here the lossless projective order is $S = X\sigma$ with $0 < \sigma \leq 1/500$. A fresh reconstruction from the original gate again gives the same twenty (S, X) terms. After the nonnegative first layer is discarded, exact centered arithmetic gives the quadratic reserve

$$c_2 \geq \frac{2564950982194530478444050838857341987999}{127401984000000000000000000000},$$

and bounds the sixteen higher terms by a rational R_{abs} whose strict difference is

$$(57) \quad m_{19} := \frac{N_{19}}{D_{19}} > 0,$$

where the exact positive integers N_{19}, D_{19} are printed in [Section B](#). The exact seam and endpoint diagnostics are also positive at all 72 falsification nodes, but those finite nodes are not used to prove the continuum. On $1/19 \leq X \leq 1/18$, the coordinate

$$u = 342(X - 1/19), \quad 0 \leq u \leq 1,$$

places the left endpoint exactly on the preceding seam. The lossless order is $S = X\sigma$ with $0 < \sigma \leq 19/10000$. Independent reconstructions of the

original gate give the same twenty quotient terms, sixteen higher terms, and 947 centered parameter monomials. After discarding only the nonnegative first layer and additional positive quadratic terms, the strict continuum reserve is

$$(58) \quad m_{18} := \frac{N_{18}}{D_{18}} > 0.$$

On the next cell $1/18 \leq X \leq 1/17$, put

$$u = 306(X - 1/18), \quad 0 \leq u \leq 1.$$

Here $0 < \sigma \leq 9/5000$, and the same independent reconstruction yields the strict reserve

$$(59) \quad m_{17} := \frac{N_{17}}{D_{17}} > 0.$$

On the next cell $1/17 \leq X \leq 1/16$, put

$$u = 272(X - 1/17), \quad 0 \leq u \leq 1.$$

Here $0 < \sigma \leq 17/10000$. A fresh source reconstruction and an independent signed- z Gram reconstruction again give the same twenty quotient terms, sixteen higher terms, and 947 centered parameter monomials. After discarding only the nonnegative first layer and additional positive quadratic terms, their common strict continuum reserve is

$$(60) \quad m_{16} := \frac{N_{16}}{D_{16}} > 0.$$

On the next cell $1/16 \leq X \leq 1/15$, put

$$u = 240(X - 1/16), \quad 0 \leq u \leq 1.$$

Here $0 < \sigma \leq 1/625$. The package source again reconstructs the literal fully conjugated Hermitian gate, while the independent referee starts from signed- z Gram columns and a direct affine-power quotient. Both recover the same twenty quotient terms, sixteen higher terms, and 947 centered parameter monomials. After discarding only the nonnegative first layer and additional positive quadratic terms, their common strict continuum reserve is

$$(61) \quad m_{15} := \frac{N_{15}}{D_{15}} > 0.$$

On the terminal cell $1/15 \leq X \leq 1/14$, put

$$u = 210(X - 1/15), \quad 0 \leq u \leq 1.$$

Here $0 < \sigma \leq 3/2000$. The package source again reconstructs the literal fully conjugated Hermitian gate, while the independent referee starts from signed- z Gram columns and a direct affine-power quotient. Both recover the same twenty quotient terms, sixteen higher terms, and 947 centered parameter monomials. After discarding only the nonnegative first layer and

additional positive quadratic terms, their common strict continuum reserve is

$$(62) \quad m_{14} := \frac{N_{14}}{D_{14}} > 0.$$

The next nine reciprocal cells are handled by the identical lossless substitution and fresh exact reconstruction. Their coordinates and sharp orders are

cell	coordinate u	projective order
$[1/14, 1/13]$	$182(X - 1/14)$	$0 < \sigma \leq 7/5000$
$[1/13, 1/12]$	$156(X - 1/13)$	$0 < \sigma \leq 13/10000$
$[1/12, 1/11]$	$132(X - 1/12)$	$0 < \sigma \leq 3/2500$
$[1/11, 1/10]$	$110(X - 1/11)$	$0 < \sigma \leq 11/10000$
$[1/10, 1/9]$	$90(X - 1/10)$	$0 < \sigma \leq 1/1000$
$[1/9, 1/8]$	$72(X - 1/9)$	$0 < \sigma \leq 9/10000$
$[1/8, 1/7]$	$56(X - 1/8)$	$0 < \sigma \leq 1/1250$
$[1/7, 1/6]$	$42(X - 1/7)$	$0 < \sigma \leq 7/10000$
$[1/6, 1/5]$	$30(X - 1/6)$	$0 < \sigma \leq 3/5000$

For each row, source and independent signed- z referee obtain the same twenty quotient terms, sixteen higher terms, and 947 centered monomials, with strict exact reserves

$$(63) \quad m_{13}, m_{12}, m_{11}, m_{10}, m_9, m_8, m_7, m_6, m_5 > 0.$$

Their full rational values are recorded in [Section B](#) and in the bound certificate notes.

On the terminal breakpoint cell use

$$(64) \quad X = \frac{1}{5} + u \left(X_* - \frac{1}{5} \right), \quad 0 \leq u \leq 1, \quad 0 < \sigma \leq \frac{1}{2000}.$$

The exact source and independently ordered direct-affine referee again reconstruct the original fully conjugated gate. The common closed-cell reserve is

$$(65) \quad m_* := \frac{N_*}{D_*} > 0.$$

Thus the gate certificate includes $X = X_*$. Equation (51), not the gate reserve, is what makes the uniform strict-danger theorem half-open.

The X16 seam at $X = 1/17$ already lies beyond the old locator $Z = 1/6$. All later cells remain beyond that obsolete locator; this causes no loss of the exact construction. The underlying compact-ball chart requires only

$$0 < S \leq 1, \quad \lambda > 0, \quad Z > 0, \quad x^2 + y^2 + Z < 1,$$

and no inverse, rank, gate, or denominator-clearing identity assumes $Z < 1/6$. The predecessor bounds and (51) verify these hypotheses on $X < X_*$. At the single zero-danger endpoint the matrix construction, rank calculation, and original gate identity remain well-defined and are checked directly; no denominator contains the danger reserve. Thus the preceding fourteen

certificate pieces and ten terminal cells prove $\widehat{Q} > 0$ throughout the closed box (43). Since $D, S > 0$, (52) proves $36\Gamma(Q_B) > 0$, and hence $\mathcal{G}(Q_B) > 0$. Notice that $S = 0$ is not in the parameterization because $\lambda = (1 + M)/S$ is then undefined. This completes the proof of Theorem 5.1.

6. A DISTINCT AFFINE- ω TILTED CHART

The centered sheet reaches its exact uniform strict-danger breakpoint by varying X while keeping the physical center of ω fixed. A separate chart can continue farther by tilting that center. Retain the ranges of S, M, ω, ν in (43), but now let

$$(66) \quad \frac{1}{5} \leq X \leq \frac{3}{13}, \quad \omega_{\text{phys}} = \omega - \frac{10636}{275} \left(X - \frac{1}{5} \right).$$

Define λ, y_0, x, y as in (45), replace ω by ω_{phys} in W_{sh} , and put

$$(67) \quad b = \frac{45M + 18}{25(1 + M)} - \frac{3}{5}X + \omega_{\text{phys}},$$

$$Z_{\text{tilt}} = 3X - X^2 + Sb - S^2y_0^2.$$

Theorem 6.1 (Affine- ω tilted continuation). *For every parameter choice in (66) and the unchanged centered ranges, choose either sign of z with $z^2 = Z_{\text{tilt}}$ and construct $Q_D = \lambda H$ from (37)–(38). Then*

$$\lambda > 0, \quad 0 < Z_{\text{tilt}} < 1, \quad 1 - x^2 - y^2 - Z_{\text{tilt}} \geq \frac{S}{100} > 0,$$

Q_D is positive semidefinite of rank two, and

$$\mathcal{G}(Q_D) > 0.$$

The chart includes both signs of z and the closed endpoint $X = 3/13$.

Proof. Both exact constructions start from the literal fully conjugated Hermitian gate. After $\lambda = (1 + M)/S$ is substituted and the positive factor $25^8(1 + M)^8S$ is cleared, the quotient has 22 (S, X) terms, 18 higher terms, and 1013 centered (M, ω, ν) monomials. Its first layer is the nonnegative quantity in (53). The lossless enlarged coordinates

$$X = \frac{1}{5} + \frac{2}{65}u, \quad S = \frac{1}{2000}\tau X, \quad 0 \leq u, \tau \leq 1$$

turn the remaining quotient into a polynomial of bidegree $(5, 7)$. Its tensor Bernstein expansion has 48 controls. Exact centered arithmetic proves all 48 strictly positive, with common minimum

$$(68) \quad R_* > 0.$$

Its exact numerator and denominator are printed in Section B. The source obtains these controls by a sparse map; the independent referee uses signed- z Gram columns, a different component order, direct affine expansion, and a dense τ -first Bernstein transform. The 72 exact seam/interior/endpoint

nodes checked by both programs are falsification diagnostics only; the 48 continuum controls prove the sign.

For legality, put

$$T = \frac{9}{5} + \omega + \frac{6}{5}\nu - \frac{63}{125(1+M)} - \frac{3}{5}X - \frac{10636}{275} \left(X - \frac{1}{5} \right).$$

Then $1 - x^2 - y^2 - Z_{\text{tilt}} = 3/5 - (13/5)X - ST$. Exact monotonic envelopes give

$$\frac{8879008598718934873}{15857127000000000000} \leq Z_{\text{tilt}} \leq \frac{8316795197}{13013000000} < 1,$$

$$T_{\max}(3/13) = -\frac{1}{100}.$$

and prove the displayed $S/100$ danger reserve. Since $\det C = (5/9)SZ_{\text{tilt}} > 0$, the compression has rank two. $\square$

At $X = 1/5$ one has $\omega_{\text{phys}} = \omega$, so this tilted chart meets the centered chart on an exact seam. Their union is therefore connected and its X -projection is $0 \leq X \leq 3/13$. The two charts have different physical centers away from the seam and must not be identified with one larger sheet. Moreover

$$\lim_{S \rightarrow 0+} (1 - x^2 - y^2 - Z_{\text{tilt}}) = \frac{3}{5} - \frac{13}{5}X < 0 \quad (X > 3/13).$$

Thus $3/13$ is an endpoint for this bounded, S -independent affine-center class, not a raw-gate negative or a full compact-ball boundary.

7. A SEPARATE INWARD- Z SHEET

The preceding obstruction concerns the affine chart's fixed leading Z -profile, not the original gate. We now move inward in Z while retaining the same affine- ω center. Keep the ranges of S, M, ω, ν from (43), let

$$(69) \quad \frac{3}{13} \leq X \leq \frac{83059}{100000}, \quad \mu = 1 + M, \quad \omega_{\text{phys}} = \omega - \frac{10636}{275} \left(X - \frac{1}{5} \right),$$

and define

$$(70) \quad \begin{aligned} \lambda &= \frac{\mu}{S}, & y_0 &= \frac{12}{25\mu} + \nu, \\ x &= -\frac{1}{5} + X, & y &= \frac{3}{5} + Sy_0, \\ b &= \frac{45M + 18}{25\mu} - \frac{3}{5}X + \omega_{\text{phys}}, & Z_{\text{in}} &= \frac{9}{13} - X^2 + Sb - S^2y_0^2. \end{aligned}$$

Equivalently,

$$(71) \quad Z_{\text{in}} = Z_{\text{tilt}} - 3 \left(X - \frac{3}{13} \right).$$

Theorem 7.1 (Inward- Z affine- ω sheet). *For every parameter choice in (69) and the unchanged centered ranges, choose either sign of z with $z^2 = Z_{\text{in}}$ and construct $Q_E = \lambda H$ from (37)–(38). Then*

$$(72) \quad \lambda > 0, \quad 0 < Z_{\text{in}} < 1, \quad 1 - x^2 - y^2 - Z_{\text{in}} \geq \frac{2}{5} \left(X - \frac{3}{13} \right) + \frac{S}{100} > 0.$$

The matrix Q_E is positive semidefinite of rank two and

$$\mathcal{G}(Q_E) > 0.$$

At $X = 3/13$, the complete tuple (λ, x, y, b, Z) agrees parameter by parameter with [Theorem 6.1](#). This is a second sheet with a seam, not one globally enlarged sheet.

Proof. First we verify legality independently of the gate sign. Exact monotonic endpoint arithmetic gives

$$(73) \quad \frac{46999}{100100} \leq y_0 \leq \frac{16333}{33300}, \quad -\frac{24601484583}{1017500000} \leq b \leq -\frac{299011}{500500} < 0$$

and

$$(74) \quad \frac{160243869095653}{15857127000000000000} \leq Z_{\text{in}} < \frac{108}{169} < 1.$$

Put $T = (6/5)y_0 + b$. Direct substitution into (70) gives the exact identity

$$(75) \quad D := 1 - x^2 - y^2 - Z_{\text{in}} = \frac{2}{5} \left(X - \frac{3}{13} \right) - ST.$$

The maximum of T at $X = 3/13$ is $-1/100$, and T decreases with X . More explicitly, the centered worst-case surplus factors as

$$D - \left\{ \frac{2}{5} \left(X - \frac{3}{13} \right) + \frac{S}{100} \right\} = \frac{10801S(13X - 3)}{3575} \geq 0.$$

This proves strict danger. Since $\mu \geq 999/1000$, one has $\lambda > 0$; the support compression satisfies

$$\det C = \frac{5}{9}SZ_{\text{in}} > 0.$$

Its leading entry is $S > 0$, so it is positive definite and Q_E has rank two. The signed variable cancels from the fully conjugated gate modulo $z^2 - Z_{\text{in}}$, so both lifts are covered.

It remains to prove the gate sign on the continuum. Let $G = 36\Gamma(Q_E)$ and clear only the displayed positive factors:

$$(76) \quad \hat{Q} = (25\mu)^8 S^3 G.$$

Definition-level expansion from Q_E, Q_E^2 has exactly 32 nonzero (S, X) coefficients and bidegree $(7, 4)$. Split

$$(77) \quad \hat{Q} = S^3 25^8 M^2 \mu^8 (5M^2 + 14M + 14) + Q_{\text{core}}.$$

The first term is nonnegative because

$$5M^2 + 14M + 14 = 5 \left(M + \frac{7}{5} \right)^2 + \frac{21}{5} > 0.$$

On the first cell $3/13 \leq X \leq 1/4$, use the lossless rectangle enlargement

It covers the physical range because $S/X \leq (1/10000)/(3/13) = 13/30000$. The exact tensor Bernstein degree of Q_{core} is $(7, 9)$, hence there are 80 controls. Centered rational arithmetic proves that the controls $(0, 0)$, $(0, 1)$, $(1, 0)$ vanish identically and that the other 77 controls are strictly positive. The least strict lower bound occurs at $(2, 0)$ and is

The three zeros are confined to the artificial $\tau = 0$, equivalently $S = 0$, closure. On the physical domain $\tau > 0$, the total Bernstein weight of the rows with first index at least two is

Indeed, when $0 < \tau < 1$ the $i = 2$ summand is positive and all summands are nonnegative, while $W(1) = 1$. Summing the u -Bernstein basis, which has total weight one, therefore gives

Combining this with (77) and dividing by the strictly positive factor in (76) proves the first cell.

$$(81) \quad X = \frac{1}{4} + \frac{u}{520}, \quad S = \frac{\tau}{10000}, \quad 0 \leq u, \tau \leq 1.$$

and its complete parameter polynomial is

Before either Bernstein transform, the cleared gate at $u = 0$ agrees term by term with the first-cell gate at $X = 1/4$.

$$(83) \quad X = \frac{131}{520} + \left(\frac{83059}{100000} - \frac{131}{520} \right) u, \quad S = \frac{\tau}{10000}.$$

This is a boundary of the displayed parameterization, not a negative value of the original gate and not a maximality statement for any other chart.

8. A RECENTERED INWARD-Z SHEET

The preceding rational bracket is an obstruction to the old formula, not to moving inward by a different amount. Put

$$(86) \quad X_0 = \frac{83059}{100000}, \quad X_{\text{rec}}^* = \frac{1019767}{1040000},$$

retain $0 < S \leq 1/10000$, $|M| \leq 1/1000$ and $|\omega|, |\nu| \leq 1/100$, and keep every definition in (70) except for the squared signed coordinate. Define

$$(87) \quad Z_{\text{rec}} = Z_{\text{in}} + 2(X - X_0).$$

Thus the added term vanishes at $X = X_0$.

Theorem 8.1 (Recentered inward-Z continuation). *For every*

$$X_0 \leq X \leq X_{\text{rec}}^*, \quad 0 < S \leq \frac{1}{10000}, \quad |M| \leq \frac{1}{1000}, \quad |\omega|, |\nu| \leq \frac{1}{100},$$

choose either sign of z with $z^2 = Z_{\text{rec}}$ and construct $Q_F = \lambda H$ from (37)–(38). Then

$$\lambda > 0, \quad 0 < Z_{\text{rec}} < 1, \quad 1 - x^2 - y^2 - Z_{\text{rec}} > 0,$$

Q_F is positive semidefinite of rank two, and

$$\mathcal{G}(Q_F) > 0.$$

At $X = X_0$ the complete parameter tuple, the cleared original gate, and the cleared certificate core agree exactly with Theorem 7.1. The right endpoint $X = X_{\text{rec}}^$ is included for every physical $S > 0$.*

Proof. We first prove legality. With $T = (6/5)y_0 + b$, direct substitution gives

$$(88) \quad D := 1 - x^2 - y^2 - Z_{\text{rec}} = \frac{2}{5} \left(X - \frac{3}{13} \right) - 2(X - X_0) - ST.$$

Moreover

$$\frac{dZ_{\text{rec}}}{dX} = 2 - 2X - \frac{10801}{275}S \geq \frac{5002423}{143000000} > 0.$$

Exact monotone endpoint bounds on the complete parameter box give

$$(89) \quad \frac{160243869095653}{15857127000000000000} \leq Z_{\text{rec}} \leq \frac{33258337711}{1081600000000} < 1.$$

The eight closed cells used below have the following strict danger bounds.

cell in X	exact lower bound for D
$[X_0, 17/20]$	135767/650000
$[17/20, 7/8]$	109767/650000
$[7/8, 9/10]$	83767/650000
$[9/10, 11/12]$	199301/1950000
$[11/12, 15/16]$	44767/650000
$[15/16, 19/20]$	31767/650000
$[19/20, 39/40]$	5767/650000
$[39/40, X_{\text{rec}}^*]$	$(4181417/143000)S$

At the terminal endpoint the S -independent part of (88) is zero, but the exact full-cell bound $T \leq -4181417/143000$ proves the last row for every $S > 0$. Finally,

$$\det C = \frac{5}{9}SZ_{\text{rec}} \geq \frac{160243869095653}{2854282860000000000}S > 0.$$

This proves compact-ball legality, rank two, and both signed lifts.

It remains to prove the gate sign. Set $G = 36\Gamma(Q_F)$ and clear the same positive factor as in (76). Definition-level expansion from the Hermitian matrices Q_F, Q_F^2 gives 34 nonzero (S, X) coefficients. After subtracting the manifestly nonnegative layer

$$S^3 25^8 M^2 (1 + M)^8 (5M^2 + 14M + 14),$$

the core again has 34 (S, X) coefficients, exactly 1659 centered monomials in (M, ω, ν) , and bidegree $(7, 4)$.

On each of the eight rows above map X affinely to $u \in [0, 1]$ and put $S = \tau/10000$, $0 \leq \tau \leq 1$. Exact centered rational arithmetic gives a tensor Bernstein polynomial with 40 lower controls. Every control is strictly positive; on every cell the unique weakest index is $(7, 0)$, with the eight exact reserves $\rho_0, \dots, \rho_7 > 0$ printed in Section B.1. Since the Bernstein basis is nonnegative and sums to one, each core is strictly positive on its closed cell. The left endpoint of every cell agrees at the parameter, cleared-gate, and cleared-core levels with the preceding right endpoint, including the seam $X = X_0$ with Theorem 7.1. Their union is the complete interval in (86), proving the theorem.

The packaged source programs construct the signed compact frame, Q_F , Q_F^2 , and the fully conjugated raw gate before applying the chart. Eight separate no-import referees reverse the Gram columns, change the cyclic component order, and reconstruct the same controls by exact dense Bernstein transforms. The 72 exact nodes per cell are retained only as falsification diagnostics. All source and referee manifests and their fail-closed attacks are included in the reproducibility archive. $\square$

The terminal endpoint is sharp only for legality of the displayed recentered formula. For $X > X_{\text{rec}}^*$ the S -independent part of (88) is negative, so sufficiently small $S > 0$ makes this chart illegal before its raw gate can be tested as a legal datum. In particular, this is neither a negative raw-gate

example nor a maximality statement for a different chart or for the full compact ball.

9. A SEAM-PRESERVING CONSTANT- Z SHEET

The obstruction just described concerns the slope of the recentered formula. It does not prevent a continuation that freezes the squared signed coordinate at the seam. Retain the parameter box and every other definition from [Theorem 8.1](#), and put

$$(90) \quad Z_{\text{const}} = Z_{\text{rec}} - 2(X - X_{\text{rec}}^*) = Z_{\text{in}} + 2(X_{\text{rec}}^* - X_0).$$

Theorem 9.1 (Seam-preserving constant- Z continuation). *For every*

$$X_{\text{rec}}^* \leq X \leq \frac{497}{500}, \quad 0 < S \leq \frac{1}{10000}, \quad |M| \leq \frac{1}{1000}, \quad |\omega|, |\nu| \leq \frac{1}{100},$$

choose either sign of z with $z^2 = Z_{\text{const}}$ and construct $Q_F = \lambda H$ from [\(37\)](#)–[\(38\)](#). Then

$$\lambda > 0, \quad 0 < Z_{\text{const}} < 1, \quad 1 - x^2 - y^2 - Z_{\text{const}} > 0,$$

Q_F is positive semidefinite of rank two, and

$$\mathcal{G}(Q_F) > 0.$$

At $X = X_{\text{rec}}^$ the complete parameter tuple, the cleared original gate, and the cleared certificate core agree exactly with [Theorem 8.1](#).*

Proof. We first establish legality on the two closed cells. With the same $T = (6/5)y_0 + b$ as in [\(88\)](#), substitution into [\(90\)](#) gives

$$(91) \quad D = \frac{2}{5} \left(X - \frac{3}{13} \right) + 2(X_0 - X_{\text{rec}}^*) - ST.$$

On $X_{\text{rec}}^* \leq X \leq 99/100$, exact monotonicity and endpoint arithmetic give

$$(92) \quad -\frac{5455801}{2750000} \leq \frac{dZ_{\text{const}}}{dX} \leq -\frac{1019767}{520000} < 0,$$

$$(93) \quad \frac{143889690210214873}{15857127000000000000} \leq Z_{\text{const}} \leq \frac{33258337711}{1081600000000} < 1,$$

$$(94) \quad T \leq -\frac{8425123367}{286000000} < 0.$$

The S -independent part of [\(91\)](#) vanishes at the left seam and is increasing. Hence

$$D \geq \frac{8425123367}{286000000} S > 0,$$

$$\det C = \frac{5}{9} S Z_{\text{const}} \geq \frac{143889690210214873}{28542828600000000000} S > 0.$$

This proves compact-ball legality, rank two, and both signed lifts.

On the adjacent cell $99/100 \leq X \leq 497/500$, the same definition-level calculation gives

$$(95) \quad Z_{\text{const}} \geq \frac{17798406223702873}{15857127000000000000} > 0, \quad Z_{\text{const}} \leq \frac{31507}{2600000} < 1,$$

$$(96) \quad \frac{dZ_{\text{const}}}{dX} \leq -\frac{99}{50} < 0, \quad T \leq -\frac{5332081}{178750} < 0,$$

$$(97) \quad D \geq \frac{9833}{2600000} > 0, \\ \det C \geq \frac{17798406223702873}{28542828600000000000} S > 0.$$

Thus the second cell is also strictly legal, rank two, and valid for both signed lifts.

For the gate sign, the source starts from the signed compact frame and the original Hermitian Q_F, Q_F^2 definitions. After the same positive clearing and removal of the displayed manifestly nonnegative layer, the cleared gate and core have 32/32 nonzero (S, X) coefficients, 1581 centered monomials, and bidegree $(7, 4)$. With X mapped affinely to $u \in [0, 1]$ and $S = \tau/10000$, exact centered rational arithmetic gives 40/40 strictly positive Bernstein lower controls on each cell. Each cell has unique weakest index $(7, 0)$, with its literal-free reserve printed in [Section B.2](#). Positivity of the Bernstein basis proves the strict gate sign on their complete connected union. The parameter, cleared-gate, and cleared-core expressions agree exactly at $X = 99/100$.

The parameter, cleared-gate, and cleared-core identities at $X = X_{\text{rec}}^*$ give the three-level seam. Two independent no-import referees reconstruct the compression from reversed signed- z Gram columns, check all nine entries of Q_F^2 , use different cyclic expansions and dense u -first Bernstein transforms, and recover the same exact certificates and legality bounds. Their 72 exact nodes per cell are falsification diagnostics only. Optimized execution, corrupted source, predecessor or manifest hashes, altered normalization, deletion of a Q^2 term, reversal of the danger sign, and deletion of a core coefficient all fail closed. $\square$

For the constant- Z chart, the exact worst-corner legality root lies strictly between $497/500$ and $199/200$. This is only a root for the displayed chart: it is neither a raw-gate negative nor a maximality statement for a different chart or the full compact ball.

10. A SEAM-PRESERVING CAP SHEAR

The constant- Z chart cannot reach $X = 1$ while retaining its old transverse coordinates: there $x_{\text{old}} = 4/5$ and $y_{\text{old}} = 3/5 + Sy_0$, so for every $S > 0$ and $Z \geq 0$,

$$(98) \quad 1 - x_{\text{old}}^2 - y_{\text{old}}^2 - Z = -S \left(\frac{6}{5} y_0 + S y_0^2 \right) - Z < 0.$$

This is a no-go only for that chart family, not a raw-gate counterexample. To move the transverse center, put

(99)

$$\delta = X - \frac{497}{500}, \quad x_{\text{cap}} = x_{\text{old}} - \frac{3}{2}\delta, \quad y_{\text{cap}} = y_{\text{old}}, \quad Z_{\text{cap}} = Z_{\text{const}} + 2\delta.$$

Theorem 10.1 (Cap x -shear continuation). *For*

$$\frac{497}{500} \leq X \leq 1, \quad 0 < S \leq \frac{1}{10000}, \quad |M| \leq \frac{1}{1000}, \quad |\omega|, |\nu| \leq \frac{1}{100},$$

choose either sign of z with $z^2 = Z_{\text{cap}}$, retain $\lambda = (1 + M)/S$, and construct $Q_H = \lambda H$ from (37)–(38) using $(x_{\text{cap}}, y_{\text{cap}}, z)$. Then Q_H is positive semidefinite of rank two, lies in the strict dangerous chart, and

$$\mathcal{G}(Q_H) > 0.$$

At $X = 497/500$ the full parameter tuple, positively cleared original gate, and cleared certificate core agree exactly with Theorem 9.1. The closed endpoint $X = 1$ is included.

Proof. Exact endpoint and monotonicity arithmetic on the complete parameter box gives

$$(100) \quad \begin{aligned} \frac{791}{1000} &\leq x_{\text{cap}} \leq \frac{397}{500}, \\ Z_{\text{cap}} &\geq \frac{17798406223702873}{15857127000000000000} > 0, \end{aligned}$$

$$(101) \quad \begin{aligned} Z_{\text{cap}}|_{X=1} &\geq \frac{17995576623934873}{15857127000000000000} > 0, \quad Z_{\text{cap}} \leq \frac{10967}{2600000} < 1, \\ T &\leq -\frac{1218219}{40625} < 0, \end{aligned}$$

$$(102) \quad D_{\text{cap}} := 1 - x_{\text{cap}}^2 - y_{\text{cap}}^2 - Z_{\text{cap}} \geq \frac{13993}{2600000} > 0.$$

Moreover $dD_{\text{cap}}/dX \geq 391/500 > 0$ and

$$\det C = \frac{5}{9}SZ_{\text{cap}} \geq \frac{17798406223702873}{28542828600000000000}S > 0.$$

This proves full-cell legality, both signed lifts, and strict rank two, including at $X = 1$.

Definition-level reconstruction from the signed compact frame, Q_H , all nine entries of Q_H^2 , and the fully conjugated raw gate gives exact degree four in Z . Positive clearing is reversible. After removing the common manifestly nonnegative layer, the cleared gate and core have 48/48 nonzero (S, X) coefficients, 2263 centered monomials, and bidegree $(7, 8)$. Mapping (S, X) to the unit square gives 72/72 strictly positive exact Bernstein controls. The unique weakest index is $(7, 8)$, with reserve recorded in Section B.3. The Bernstein basis therefore proves the gate sign on the full continuum. The 72 exact raw-gate nodes are retained only as diagnostics.

At $X = 497/500$ the increment in (99) vanishes, giving the three-level seam. An independent literal-free no-import referee reverses the signed Gram columns, changes the Q^2 multiplication and elimination order, and derives the 72 controls without reading candidate coefficients. Its normal run passes and eight source, manifest, predecessor, normalization, deleted-square, flipped-danger, deleted-core, and optimized-mode attacks fail closed. $\square$

11. AN ALL-SCALE STITCHED TWO-COORDINATE LOCAL BOX

The next family is independent of the stitched X -sheet and opens two transverse shape coordinates at fixed $Z = 1/8$. It is itself the exact union of four adjacent rational cells, not one unchecked extrapolation.

Theorem 11.1 (All-scale stitched $Z = 1/8$ local box). *Assume*

$$(103) \quad \begin{aligned} 0 < h \leq 1, \quad \lambda > 0, \quad Z = z^2 = \frac{1}{8}, \\ \frac{313}{500} \leq x \leq \frac{7}{10}, \quad |y| \leq \frac{1}{100}. \end{aligned}$$

For either sign of z , construct $Q_I = \lambda H$ from (37)–(38). Then Q_I is positive semidefinite of rank two, lies in the strict dangerous chart, and

$$\mathcal{G}(Q_I) > 0.$$

Proof. Put $S = h^2$ and partition the x interval into

$$\left[\frac{313}{500}, \frac{627}{1000} \right], \quad \left[\frac{627}{1000}, \frac{63}{100} \right], \quad \left[\frac{63}{100}, \frac{16}{25} \right], \quad \left[\frac{16}{25}, \frac{7}{10} \right].$$

On the whole stitched box,

$$(104) \quad D = 1 - x^2 - y^2 - Z \geq 1 - \left(\frac{7}{10} \right)^2 - \left(\frac{1}{100} \right)^2 - \frac{1}{8} = \frac{3849}{10000} > 0.$$

The signed Gram construction gives the same Hermitian compression for both signs of z , and

$$\det C = \frac{5}{72} S > 0, \quad \operatorname{Re}(Q_{IPB})_1 = -\frac{5\sqrt{6}}{36} \lambda S D < 0.$$

Thus the entire box is strictly legal and rank two.

Direct reconstruction of the fully conjugated gate from Q_I, Q_I^2 gives

$$(105) \quad 36\Gamma(Q_I) = \sum_{m=0}^4 C_m(S, x, y) \lambda^m, \quad C_0 = 5, \quad S \mid C_m \ (m \geq 1).$$

Each cell is mapped independently to the unit cube, with $y = -1/100 + u/50$ and its own affine rational x map. The exact three-variable power-to-Bernstein conversions yield

cell	D lower bound	controls	left seam	unique weakest control
[313/500, 627/1000]	481771/1000000	531/531	71/71	C_3
[627/1000, 63/100]	239/500	531/531	71/71	C_3
[63/100, 16/25]	4653/10000	531/531	71/71	C_3
[16/25, 7/10]	3849/10000	531/531	71/71	$C_1(0, 2, 2) = 517/1000$

On each cell the five coefficient groups have degrees and control counts

degree	(0, 0, 0)	(0, 2, 2)	(1, 4, 4)	(2, 6, 6)	(3, 8, 8)
controls	1	9	50	147	324.

All 531/531 controls are strictly positive. Since $S > 0$ and $\lambda > 0$, tensor Bernstein positivity proves (105) strictly positive on every closed cell.

At $x = 313/500, 627/1000, 63/100, 16/25$, exact reconstruction verifies the parameter, Hermitian compression, fully conjugated raw gate, and all 71 boundary controls. Thus the four cells splice without a gap. The source programs derive the frame, C, H, Q_I, Q_I^2 , and raw gate before their cell maps. Their 81 exact nodes per cell are diagnostics only. Four independent literal-free no-import referees change the signed-Gram, Q_I^2 , elimination, and Bernstein-transform orders and reconstruct every continuum certificate and seam. The latest cell has $D \geq 3849/10000$, $\det(C)/S = 5/72$, and unique weakest control $C_{1,(0,2,2)} = 517/1000$. No sampled sign is used as proof. $\square$

12. AN ALL-SCALE HIGH- Z COMPLEX-PHASE TUBE

We retain the compact-ball chart (37)–(38), including the normalization $p = p_B$. The following family is geometrically independent of the moving sheet in Theorem 5.1: its displacement covers the complete interval $0 < S \leq 1$, its positive scale is arbitrary, and its shape lies near the upper spherical boundary.

Theorem 12.1 (High- Z complex-phase tube). *Assume*
(106)

$$0 < S \leq 1, \quad \lambda > 0, \quad 0 \leq x \leq \frac{1}{8}, \quad |y| \leq \frac{1}{64}, \quad Z = z^2 = \frac{63}{64} - x^2.$$

Choose either sign of z , put $h = \sqrt{S}$, and construct $Q_C = \lambda H$ from (37)–(38). Then Q_C is positive semidefinite of rank two, lies in the strict dangerous chart, and

$$\mathcal{G}(Q_C) > 0.$$

The statement includes all finite positive scales λ and both signs of z ; it does not include the artificial closure $S = 0$ or projective scale infinity.

Proof. First,

$$(107) \quad 1 - x^2 - y^2 - Z = \frac{1}{64} - y^2 \geq \frac{63}{4096} > 0, \quad \frac{31}{32} \leq Z \leq \frac{63}{64}.$$

Thus the family is legal and strictly dangerous by (39). The support compression in (37) is Hermitian, has positive leading entry S , and has determinant

$$\det C = \frac{5}{9}SZ > 0.$$

Consequently Q_C is positive semidefinite of rank two. The compact chart depends on signed z only through $Z = z^2$, so this argument and the gate calculation below cover both signs.

Reconstruct the fully conjugated gate (12) directly from Q_C and Q_C^2 . Exact reduction by $S = h^2$ gives

$$(108) \quad 36\Gamma(Q_C) = 5 + \sum_{m=1}^4 C_m(S, x, y, Z)\lambda^m, \quad C_m = SG_m.$$

Map (106) losslessly to a unit cube by

$$(109) \quad u = 8x, \quad y = \frac{2v-1}{64}, \quad Z = \frac{63-u^2}{64}, \quad 0 \leq u, v \leq 1.$$

The exact tensor Bernstein data for G_m on $(S, u, v) \in [0, 1]^3$ are as follows.

polynomial	degree (S, u, v)	controls	least control; index
G_1	$(0, 1, 2)$	6	33605/6144; $(0, 1, 2)$
G_2	$(1, 2, 4)$	30	15241448155/1358954496; $(0, 2, 4)$
G_3	$(2, 3, 6)$	84	49463834927165/8349416423424; $(0, 3, 6)$
G_4	$(3, 4, 8)$	180	1152452992382662513/307792887033102336; $(0, 4, 8)$

All 300 controls are strictly positive. By Lemma 13.1, $G_m > 0$ on the closed shape cube. Since $S > 0$ and $\lambda > 0$, (108) yields

$$36\Gamma(Q_C) = 5 + S \sum_{m=1}^4 G_m \lambda^m > 5 > 0,$$

and hence $\mathcal{G}(Q_C) > 0$.

For a scale-compact audit only, put $\tau = \lambda/(1 + \lambda)$. The natural tensor degree of $(1 - \tau)^4 36\Gamma(Q_C)$ in (S, u, v, τ) is $(4, 4, 8, 4)$. Its 1125 exact controls consist of 945 positive controls and precisely 180 zeros, namely

$$(i_S, i_u, i_v, i_\tau) = (0, i_u, i_v, i_\tau), \quad 0 \leq i_u \leq 4, \quad 0 \leq i_v \leq 8, \quad 1 \leq i_\tau \leq 4.$$

Thus every zero is confined to the artificial $S = 0$ closure; no negative control or other zero occurs. The least positive full control is 33605/98304. An independent exact referee reconstructs the signed- z matrix gate and the same tensors by rational nodal collocation rather than by importing the source polynomial or using power-to-Bernstein conversion. $\square$

Remark 12.2 (Strict containment of the bridge arc). The center section $y = 0$ is the previously independently audited high- Z bridge arc. The domain (106) strictly contains that one-dimensional shape arc because it also includes every $0 < |y| \leq 1/64$. Its face $x = 1/8$ lies on the earlier center-box boundary $Z = 31/32$. These intersections do not cover either axis cap: the theorem

fixes the single surface $Z = 63/64 - x^2$ and opens it in only one phase direction. In particular, it is not a full compact-ball theorem.

13. EXACT CERTIFICATE METHODS AND INDEPENDENT AUDITS

We record the elementary certificate principles used above so that no floating-point inference is hidden in the proofs.

Lemma 13.1 (Tensor Bernstein positivity). *Map a rational box affinely to $[0, 1]^d$. If a polynomial has tensor Bernstein expansion*

$$f(x) = \sum_{\alpha} b_{\alpha} \prod_{j=1}^d \binom{n_j}{\alpha_j} x_j^{\alpha_j} (1 - x_j)^{n_j - \alpha_j}$$

and every rational control b_{α} is positive, then $f > 0$ throughout the box.

Proof. The tensor Bernstein basis functions are nonnegative and sum to one. Thus f is a convex combination of the controls at each point. $\square$

Lemma 13.2 (Exact derivative bridge). *Let f be a polynomial on $K \times [t_0 - \delta, t_0 + \delta]$. If $f(\cdot, t_0) \geq m$ on K and an exact monomial estimate gives $|\partial_t f| \leq M$ on a containing box, then*

$$f(\cdot, t) \geq m - \delta M.$$

Proof. Apply the one-variable mean-value theorem on each line with the remaining coordinates fixed. The monomial estimate is valid because each monomial is bounded by the product of the exact coordinate absolute bounds. $\square$

Lemma 13.3 (Exact nodal reconstruction). *If a polynomial has degree at most d_j in coordinate j , then its values on a Cartesian product of $d_j + 1$ distinct rational nodes determine all coefficients exactly. In particular, exact Vandermonde inversion can certify both the claimed degree and every coefficient.*

Proof. The multivariate evaluation matrix is a Kronecker product of univariate Vandermonde matrices. Each factor is invertible over $\mathbb{Q}$ because its nodes are distinct. $\square$

For [Theorem 4.1](#), the source computation converts power coefficients directly to Bernstein controls. The isolated referee reconstructs the cubic from [\(12\)](#), evaluates exact rational tensor nodes, and inverts the five Bernstein collocation axes. The t -layer source and referee independently reconstruct all seven theorem-relevant derivative majorants and the complete terminal dependency graph.

For [Theorem 5.1](#), the source computation begins with the original Q, Q^2 gate and performs the sparse moving-map expansion. The predecessor referee instead constructs signed- z Cholesky/Gram columns and performs a 72-node rational Vandermonde inversion at safe degree $(7, 8)$. All later

adjacent-cell and breakpoint referees separately reconstruct both the original Hermitian gate and its Gram-vector quartic, regenerate the twenty coefficients and all 947 centered parameter monomials, and verify every closed stitch. On the cross-cap cell, exact witnesses have opposite signs of $Z - 1/8$ and positive raw gates. Direct use of the old outer box is impossible because $y - 1/2 = 1/10 + Sy_0 > 1/10$ throughout the actual family; this is a coordinate coverage gap, not a counterexample.

For [Theorem 6.1](#), the source uses a sparse quotient and nested Bernstein conversion, while the independent signed- z referee uses direct affine expansion and dense rational collocation. They agree on all 48 controls and on the complete legality data. None of these proofs accepts a floating-point sign, a sampled minimum, or a cached discovery polynomial.

For [Theorem 7.1](#), the source starts from the signed- z compact frame and reconstructs the fully conjugated Q, Q^2 gate and a reordered Gram-vector formula before applying the sheet map. The no-import referee reverses the two Gram columns, uses the cyclic component order (third, first, second), and performs a dense u -first, τ -second Bernstein conversion. It independently recovers the 32/32/1581 coefficient counts, all 80 controls, the three artificial closure zeros, the 77 strict controls, R_*^{in} , and (80). Its 72 rational nodes are diagnostics, not a positivity inference. For the two adjacent cells, frozen source and a definition-level no-import referee again begin at the fully conjugated raw gate, but use different elimination and Bernstein-transform orders. They independently recover the two 40/40 strict tensors, the reserves (82)–(84), both splices, both signed lifts, full-cell legality, and the rational chart-root bracket. The referee uses the candidate source bytes only to verify its SHA-256 digest. Normal execution, optimized-mode rejection, corrupted-hash, deleted-square, flipped-danger, and deleted-core attacks all fail closed as intended.

For [Theorem 8.1](#), each source reconstructs the signed- z compact frame and original Hermitian Q, Q^2 gate before the new Z map is applied. Its paired no-import referee uses reversed Gram columns, a different cyclic component order, and a dense Bernstein transform. All eight pairs independently recover the 34/34/1659 structure, bidegree (7, 4), every one of the 40/40 strict controls, the complete unique weakest polynomial, three-level stitches, both signed lifts, and full-cell legality. The final pair independently derives X_{rec}^* and proves the S -dependent danger bound at the endpoint. Optimized mode, bad hashes, deleted Q^2 terms, flipped danger, and deleted-core mutations fail closed.

For [Theorem 9.1](#), the two frozen sources bind the terminal recentered predecessor and reconstructs the signed frame, Hermitian Q, Q^2 , degree-four Z dependence, positive clearing, and all three seam layers before testing legality or controls. The independent no-import referees treat the sources only as opaque hash-bound byte strings, reverse the signed- z Gram columns, eliminates in a different order, and recovers the 32/32/1581 structure, bidegree (7, 4), both sets of 40/40 strict controls, both unique weakest polynomials,

and the full-cell bounds (92)–(97). Normal and byte-compiled runs pass; the optimized, hash-corruption, normalization, deleted-square, flipped-danger, and deleted-core attacks fail closed at their intended gates.

For Theorem 10.1, the source first proves the old-family endpoint no-go (98), then reconstructs the new signed frame and the original Q, Q^2 gate before applying the shear. Its no-import referee derives independently the three-level seam, 48/48/2263 structure, bidegree (7, 8), all 72/72 strict controls, and the closed-endpoint legality bounds (100)–(102). The source bytes are used only as an opaque hash binding.

For Theorem 11.1, four source/referee pairs independently derive all five λ -coefficient polynomials directly from the fully conjugated gate. They use different signed-Gram, Q^2 , elimination, and Bernstein-transform orders, yet recover the same five tensor shapes and all 531/531 strict controls on every cell. Each left face has a complete 71/71 exact parameter/Hermitian/raw-gate/control seam. The source’s 81 rational nodes per cell are diagnostics only. Deleted-control, broken-seam, omitted-sign, and corrupted normalization attacks fail closed.

For Theorem 12.1, the source reconstructs the orthonormal frame, Hermitian compression, Q , and Q^2 . It also rebuilds every conjugated coordinate in (12), then converts exact power coefficients to the 300 Bernstein controls. The independent referee starts with a signed variable z and two Gram atoms, multiplies Q^2 directly, eliminates the unit-circle relation, and obtains every control by exact rational collocation. Both programs prove the $945 + 180$ full-tensor classification, all boundary faces, and the two-sign statement. Both fail closed under optimized Python, a corrupted dependency hash, and deletion of a raw C_4 monomial. The deleted-term residue is exactly $25S^4\lambda^4x^4/81$.

14. SCOPE AND UNRESOLVED PROBLEM

The nine positivity theorems establish strict positivity only on the displayed families. They leave open all of the following:

- arbitrary complex Hermitian Q in the balanced rank-one-side scalar gate;
- the complementary feasible-center branch outside the stated families;
- either complete high- Z axis cap, a full positive- Z collar, or the full compact unit ball;
- any identification of the centered and affine- ω tilted charts away from their seam, or any covering result beyond their connected union;
- any identification of the affine- ω and inward- Z charts away from their $X = 3/13$ seam;
- any identification of the inward- Z and recentered- Z charts away from their $X = 83059/100000$ seam, of the recentered- Z and constant- Z charts away from their $X = 1019767/1040000$ seam, or of the constant- Z and cap-shear charts away from their $X = 497/500$ seam;

- any claim that closure of this one stitched X -sheet through $X = 1$ covers the surrounding compact ball or yields a global parameterization;
- any enlargement of the stitched $Z = 1/8$ local box beyond $313/500 \leq x \leq 7/10$, or to arbitrary Z , arbitrary transverse shapes, or the full compact ball;
- arbitrary normalized scale in the moving-sheet chart;
- unbalanced rank-two densities, cases where neither positive side has rank one, higher positive ranks, and arbitrary dimension;
- the existence of one common metric for all separators;
- arbitrary-node operator bridges and the optimal constant of a fixed crossing lens.

The scale theorem contains no equality in its legal family. The moving-sheet gate is strict on $0 \leq X \leq X_*$, but uniform strict danger holds only for $X < X_*$; the unique endpoint corner (47) has zero danger. The number X_* is therefore sharp only for uniform legality on this fixed centered five-parameter box, not for the raw gate sign. In particular, $X = 1/4$ is nonlegal at a centered corner and supplies no gate counterexample. The inward- Z gate is strict for $S > 0$; its three zero Bernstein controls on the first cell occur only on the artificial $S = 0$ closure, while both later cells have 40/40 strict controls. The rational root bracket records only loss of the old inward chart's legality. The recentered sheet has eight 40/40 strict cells and remains physically legal at its exact danger-root endpoint because $S > 0$; its obstruction immediately to the right is again chart illegality, not a raw-gate sign. The constant- Z sheet has two 40/40 strict cells; the chart-legality root beyond $497/500$ is not a raw-gate sign or a maximality claim. The cap shear has 72/72 strict controls and reaches $X = 1$, but only inside its displayed five-parameter chart. The $Z = 1/8$ local box is the union of four exact cells, each with 531/531 strict controls; it opens only the two displayed shape coordinates and stops at $x = 7/10$ without a maximality claim. The high- Z tube gate is strict for $S > 0$. The face $S = 0$ is used only as an algebraic closure for coefficient extraction and is not part of either theorem domain. The tube covers one two-real-dimensional surface family, not a three-dimensional cap.

REPRODUCIBILITY AND DISCLOSURE

All proof decisions in the nine computational positivity theorems use exact rational or algebraic arithmetic. The source and isolated referee programs, audit notes, and artifact manifests are listed in the accompanying `README.md`. The referee programs import neither discovery code nor the source verifier, and they fail closed under optimized Python execution. The frozen v7 baseline already binds exact replays of the 50/919 and X17 source and independent-referee programs together with all predecessor certificates. For the frozen v10 candidate, the phase-tube and X16 source verifiers and

their independently constructed referees were also replayed, and their direct manifests were checked. For the frozen v13 candidate, the X13 through X5 reciprocal-cell sources and the X4 breakpoint source, together with ten independently constructed referees, were replayed and their direct and transitive manifests were checked. The packaged X18, X16, X15, X14, X13 through X5, and X4 source and referee copies derive their dependency root from their own file location, confine resolved dependencies to the release-local certificate workspace, and fail closed outside the supported release layouts. The X5 audit history records a stale-results integrity repair, and the X4 audit history records two pre-freeze SymPy normalization repairs; both were verifier-audit events before the frozen artifacts and neither was a mathematical failure. The frozen v12 manuscript/archive has its own independent final audit. The frozen v13 package added the first inward- Z source, no-import referee, test record, and manifests and subsequently passed an independent final manuscript/archive audit with no fatal or major finding. The frozen v15 corresponding-author revision added the endpoint and legality-frontier source/referee pairs, their exact test records, and both manifest chains; it also repaired the v13 README's incorrect line-oriented checksum command for its JSON release manifest. The frozen v16 release adds the eight recentered-sheet source/referee pairs and the constant- Z source/referee pair, their exact test records, audit notes, and eighteen manifest chains. The frozen v17 release adds the second constant- Z cell, the cap-shear theorem, and the independent $Z = 1/8$ local box, together with their source/referee pairs, exact records, audit reports, and manifest chains. Its repaired candidate passed the independent manuscript/archive reaudit. This v18 candidate upgrades that local-box theorem by the four independently audited adjacent positive- x cells through $x = 7/10$ and adds their complete frozen source/referee evidence. It is not promoted to submission-ready status until a fresh independent manuscript/archive audit has passed.

Symbolic computation was used to generate and verify the exact certificates. An AI coding assistant copied the frozen layout, transcribed the previously certified endpoint changes and the independently audited phase-tube, inward- Z , recentered- Z , constant- Z , cap-shear, and stitched local-box theorems, updated the reproducibility package, and assisted with typesetting and build checks. No Lean or other proof assistant formalizes the complex theorems in this paper; existing formalizations elsewhere in the project concern narrower real algebraic cores and are not cited as support here. Any final disclosure language, acknowledgments, and funding statement must be confirmed against the target venue's policy before submission.

APPENDIX A. PROOF OF THE ARROWHEAD ESTIMATE

Unitary conjugation diagonalizes H_0 , and diagonal phases make the two entries of y nonnegative. Thus it is enough, by continuity, to consider

$$\mathcal{A} = \begin{pmatrix} a & 0 & u \\ 0 & b & v \\ u & v & 0 \end{pmatrix}, \quad a, b \in \mathbb{R}, \quad u, v \geq 0,$$

away from multiple eigenvalues. If $a, b \geq 0$, write the unique negative eigenvalue as $-x$, $x > 0$. Its characteristic equation is

$$x(a+x)(b+x) = u^2(b+x) + v^2(a+x).$$

It follows that $x(a+b+x) \geq u^2 + v^2$, and hence

$$\|\mathcal{A}\|_1^2 - (a+b)^2 - 4(u^2 + v^2) \geq 0.$$

The case $a, b \leq 0$ follows by replacing $\mathcal{A}$ by $-\mathcal{A}$.

For the mixed-sign case write $a \geq 0$, $b = -c \leq 0$. If $\det \mathcal{A} = cu^2 - av^2 \leq 0$, write the negative eigenvalue as $-x$, $x \geq c$. Exact subtraction from the characteristic equation gives

$$x\{(a+x)(x-c) - u^2 - v^2\} = av^2 - cu^2 \geq 0.$$

Since $\|\mathcal{A}\|_1 = a - c + 2x$, the desired estimate follows. If $\det \mathcal{A} \geq 0$, write the positive eigenvalue as $z \geq a$; then

$$z\{(z-a)(z+c) - u^2 - v^2\} = cu^2 - av^2 \geq 0,$$

and $\|\mathcal{A}\|_1 = 2z - a + c$ gives the same conclusion. Degenerate cases follow by continuity, proving Lemma 2.2.

APPENDIX B. SELECTED EXACT CERTIFICATE DATA

For the t -derivative bridge in Theorem 4.1, all fourteen freshly recomputed reserves $m_f - (50/919)M_f$ are listed below in the order danger, $\Delta, n, C_0, C_1, N_0, N_1$. When a numerator is wrapped across two lines, the digit strings are to be concatenated without a space.

low r-cell:

93/500

12809761/4595000

655745491/2297500

242925468045042215653163747409/8916041034012500000000000000

536600575672403488016095079753/8916041034012500000000000000

271024413499730373992194599135992059116129594487

/24096449701525219240000000000000000000000000000

527592607098284468088362545547608061

/7132832827210000000000000000000000000000000000

high r-cell:

87/500

13177361/4595000

655791441/2297500

262886764340622837232304997121/8916041034012500000000000000
 583519690684298422184716005553/8916041034012500000000000000
 695222565537315762990076093059939765373235563737587
 /24096449701525219240000000000000000000000000000
 595031495123164277277188869161948311
 /7132832827210000000000000000000000000000000000

Every numerator and denominator is a positive integer. The literal smallest reserve is the high-cell danger value $87/500$; among signs with nonzero t -derivative, the unique finite ratio bottleneck is the low-cell N_0 . The independent 96-corner attack (48 cases with $g > 0$ and 48 with $g \leq 0$) has minimum direct $T = 1$ gate

$$\frac{40614857500397194448289907112063}{172677734175446220361000000000} > 0.$$

This corner value is a diagnostic, whereas the fourteen reserves prove the continuum.

For [Theorem 5.1](#), the five exact higher-layer bounds are:

B1 = 1040210347071333285995390774300441067433076882959
 /171992678400
 B2 = 513625292315565411172470351664061881087569457669
 /286654464000
 B3 = 9574969116015939526109085821350045427090716403
 /6370099200
 B4 = 34218452213929153632941920549574435172203
 /1592524800
 B5 = 1012066220495792924792495220066012001
 /5308416000

The exact projective margins are listed next; wrapped digit strings are to be concatenated without a space:

m_CO numerator:
 2157675883946338861464573804693908853416834606352
 657354865271
 m_CO denominator:
 10749542400
 m_C1 numerator:
 8936132140375249528801151592509689511386616532366
 8871164167
 m_C1 denominator:
 4478976000
 m_adj numerator:
 4295035787646382703962747769216564803855644527691
 1579
 m_adj denominator:
 9848008721616076800000000000000000000000000000000
 m_next numerator:
 345803391420521969132794684450316715874438964711696895
 50375704523100547
 m_next denominator:
 1719926784000
 00000

[illegible]

These displayed rationals are summaries of the independently reconstructed exact certificates, not decimal approximations. On the X14 falsification grid, at $S = 1/10000$, $M = -1/1000$, and $\omega = \nu = -1/100$, the seam and right endpoint respectively give

[illegible]

These endpoint evaluations and every 72-node grid are falsification checks only; the rational margins $m_{14}, \dots, m_5, m_*$ prove the successive closed continua. The affine- ω chart uses the different exact reserve R_* in (68), reconstructed from all 48 strict Bernstein controls.

B.1. Recentered inward- Z reserves. For the eight consecutive cells in [Theorem 8.1](#), the exact unique weakest lower controls $\rho_0, \dots, \rho_7$ are the following positive rationals. Wrapped digit strings are to be concatenated without a space.

```
rho_0 numerator:
498986825342804290410045886737376873289010315560072950081729284640738346
2088295685657449907
rho_0 denominator:
374587155000115200000000000000000000000000000000000000000000000000000000
00000000

rho_1 numerator:
363964007553160826089535269163561090689140669958074025207807180716683038
70073911227
rho_1 denominator:
239735779200073728000000000000000000000000000000000000000000000000000000

rho_2 numerator:
427041414684728450062105547867326110150274820063534048090939139748548122
01090200227
rho_2 denominator:
239735779200073728000000000000000000000000000000000000000000000000000000

rho_3 numerator:
496760245305561992027317667355427017525227533684414819777515116506941155
71509875227
rho_3 denominator:
```

For [Theorem 11.1](#), every cell has control counts 1, 9, 50, 147, 324, totaling 531. The four exact cell summaries are

cell	D reserve	seam	unique weakest value
[313/500, 627/1000]	481771/1000000	71/71	130268743001281/237304687500000
[627/1000, 63/100]	239/500	71/71	533736176784030589/972000000000000000
[63/100, 16/25]	4653/10000	71/71	66774911/121500000
[16/25, 7/10]	3849/10000	71/71	517/1000

In every row $\det(C)/S = 5/72$. The final minimum is $C_{1,(0,2,2)} = 517/1000$; the first three minima belong to the C_3 group. The complete source-note-results-report path and SHA-256 index is `certificates/xy_box_post_v17/INDEX.md`. The four source/source-manifest/referee/referee-manifest tuples are:

```
6ce356083b31927a41808ec54078e104358db2dcdce421cd3ccfc5dfaedef6c92
2ce9ef8ed0dface0c00328a65be9dc7c4c4a1f7c8cfb1053ac21ea96967ff15a
a74ab0e0e9cadce6baf07add966d3f1dc9af83ba697dd67f9d15aacced3cc07d
089306b1a1bc5cb7e07f4638dcf3f29c25ae377ae4a9034d1b631f9173f5bef3
823fbb26a31378530567674cce08ec7f7203c9882256dabed2a776e2ae6b7825
ac8c581ef64e6de4b03e657f3c9201d2864a86d3ee0af91021607a3613c13f2e
416e35a8a75762771fa7e6a271c29bc864ea640a386c5da888e2c4119a3400ce
e036d4a6bdb2625a2ad554b1a4636d4fa55c0e1f79a19d0635c3cb8c6bac62b7
da611b4325ce935c6e9857a2867d92a51a3426a07cb4a1ae1db4f12bc53088cc
dd0d83483d0b8f8c3e6d4f2626dcf89398ff992b5900463811eef70016aa0022
00b03875230c5b45729e66f95a880668ca7eb3e29814556aa91a22088d3c94a2
f3b581e6445d9c88c161a7ee25ba45a620157ab4b59805ea2328017bbf5b55df
2e646d18b9d40ebb286a59e455176709a4306401a992fb6638e5552dde90599
3badb8170b102316c2890131fba7b7061a6c5d48fb0d96979fa4dc2c50278a05
21a3d4929a91ed285a23364a93bce55fcae34f78021b99d982b24c40ef5c3d0f
ab497254f905795fc30894d7967217aa4b811e9392e2d498390f19f1920413c6
```

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